

Machine Learning

CSE 343/543 ECE 363/563

Saket Anand



INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI



Today's Class



- People Involved
- Is Machine Learning affecting us?
- What is Machine Learning?
- A brief History
- Example Systems that use Machine Learning
- Related areas
- What should you learn?
- Course Outline
- Administritivia



People Involved



- Instructor: Saket Anand

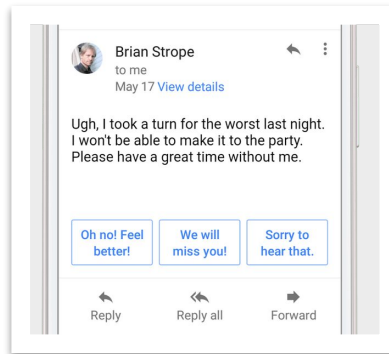
- Email: anands@iiitd.ac.in; Phone: +91 11 2690 7425; Campus Ext: 425
- Lectures: Tuesdays & Thursdays, 4:30PM–6:00PM, C101, LHC.
- Office Hours:
 - Time: Tuesdays, 12:30PM–1:30PM, or by appointment
 - Venue: B-410, R&D Block
- Course Page: <https://classroom.google.com/c/ODU1ODE0MTAwNTQ5>
- Discussions/Submissions: [Google Classroom](#) (Class code: **pnzakyya**)

- Teaching Assistants

- Ayaan Haider (PhD), Pradyuman Thakur (PhD), Parth Goyal (B. Tech.), Garvit (B. Tech.), Adamyia Rangi (B. Tech.), and more TBD
- Tutorials: Fridays, 2:00PM – 4:00PM, C101, LHC

Is Machine Learning Really Affecting Us?

- Gmail and Google Search
 - Smart Reply, Personalized Ads



- Social Media Feeds
 - Facebook, LinkedIn, Instagram
- Music and Media Streaming
 - Netflix, Youtube, Spotify

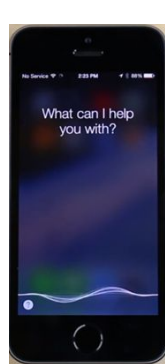
- Maps, Navigation and Travel
 - Google, Uber, Ola
- Banking and Finance
 - IVR, Fraud Protection

- Online Shopping
 - Amazon, Flipkart

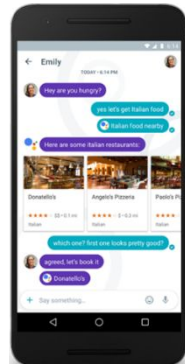


Natural and Spoken Language Processing

Conversational Agents and Translators



Apple
Siri



Google
Assistant



Microsoft
Cortana



Optical Character Recognition and Translation



Screenshot from the movie Cars 3 on Netflix India

Image Analysis and Computer Vision



Google Cloud
Vision

Shot on iPhone Xs



Shot on Pixel 3



iPhone Xs vs. Pixel 3's Night Sight Mode

Source: [XDA Dev. review](#)

Real World Google Vision API Results

A collage of three images with overlaid Google Vision API labels. The top image shows a car stereo with labels for 'technology', 'motor vehicle', 'electronics', 'stereophonic sound', 'automotive design', 'vehicle audio', 'radio', 'media player', 'audio', and 'electronic device'. The bottom-left image shows a red car tail light with labels for 'red', 'light', 'motor vehicle', 'automotive lighting', 'automotive design', 'auto part', 'automotive tail & brake light', 'car', and 'automotive exterior'. The bottom-right image shows a woman wearing a yellow hard hat and glasses with labels for 'glasses', 'eyewear', 'yellow', 'vision care', 'sunglasses', 'fashion accessory', 'smile', 'headgear', 'hard hat', and 'girl'. A large Google Cloud Vision logo is in the bottom right corner.

technology : 90.4%
motor vehicle : 88.7%
electronics : 82.9%
stereophonic sound : 77.3%
automotive design : 75.3%
vehicle audio : 74.3%
radio : 69.6%
media player : 66.5%
audio : 64.9%
electronic device : 63.1%

red : 97.0%
light : 82.4%
motor vehicle : 82.2%
automotive lighting : 80.3%
automotive design : 74.3%
auto part : 66.1%
automotive tail & brake light : 65.8%
car : 61.6%
automotive exterior : 58.1%

glasses : 97.9%
eyewear : 96.3%
yellow : 95.1%
vision care : 93.8%
sunglasses : 89.1%
fashion accessory : 85.6%
smile : 83.9%
headgear : 82.8%
hard hat : 78.5%
girl : 77.0%

Presented by Lowell Cady

Autonomous Driving



Volvo's driverless truck



Mercedes-Benz



Volkswagen



Autonomous Driving



AI first Robotics @Skild.AI



Learning by Watching Humans @Skild.AI



Pitfalls and Perils of Machine Learning

- Chinese billionaire's face identified as jaywalker
 - Surveillance system misinterpreted a billboard image for a jaywalker



- Google's panorama generation thought a skier was a mountain!

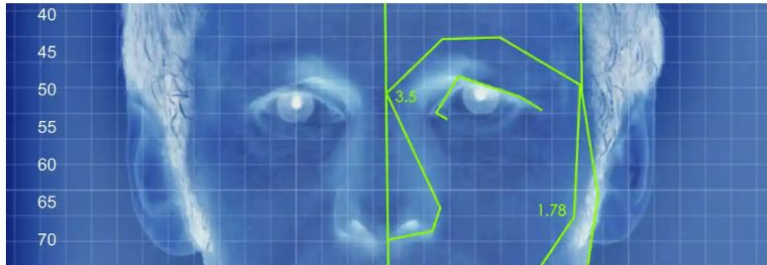


Pitfalls and Perils of ML

- Claim: Machines have a better “gaydar”
 - If true: privacy breach
 - If false: reinforces prejudice

New AI can guess whether you're gay or straight from a photograph

An algorithm deduced the sexuality of people on a dating site with up to 91% accuracy, raising tricky ethical questions



¹Source: [The Guardian](#)

- Uber hits and kills a pedestrian in Tempe, Arizona, US



Source: [the Medium](#)

- “DeepFakes” used to generate fake porn using Hollywood celebrity facial images

Efforts towards AI for Social Good

- Microsoft AI for Earth



Microsoft

- Climate Change, Agriculture, Biodiversity and Water

- Google AI for Social Good

- Flood prediction
- Cardiac risk prediction
- Mapping global fishing activity



Google AI

- Wadhvani Institute for AI



- Focus on Societal problems in India: Health, Agriculture, Education, Financial Inclusion

- Center for AI in Society, USC



- Focus on public health, social work, conservation, safety and security

- Many others...

What is Social Good?

- Inspired by UN's SDGs
 - Public Health
 - Education
 - Energy
 - Food Security
 - Conservation
 - Environment
 - Assistive Technologies
 - Mobility, and more...
- Positive Social change
 - Technology that caters to the needs of underserved communities
 - Healthcare / education / information in rural populations
 - Conservation of natural habitat for wildlife
 - Urban mobility - vulnerable road users



AI for Social Good

- Stanford University @ NeurIPS `21
 - Earth Sciences & CSE
- 15 benchmark tasks across 7 SDGs,
 - economic development, agriculture, health, education, water and sanitation, climate action, and life on land.



Tackling Climate Change with Machine Learning

	Computer vision	NLP	Time-series analysis	Unsupervised learning	RL & Control	Causal inference	Uncertainty quantification	Transfer learning	Interpretable ML	Other
Electricity Systems	1	1.1	1.1 1.2	1	1.1		1.1 1.2	1.3	1.1	1.1
Transportation	2.1 2.2 2.4		2	2.1 2.4	2	2.1 2.4	2	2.1 2.4	2	
Buildings & Cities	3.2	3.3	3	3	3.1	3.1	3.3	3		
Industry	4.1 4.3		4.3	4.3	4	4.2 4.3		4.2 4.3	4.3	
Farms & Forests	5.1 5.3 5.4				5.2			5.4		
CO ₂ Removal			6.3				6.3	6.3		6.2
Climate Prediction	7.1		7				7.3		7	
Societal Impacts	8.1 8.4	8.4	8.2 8.3		8.2	8.3	8.2	8.1	8.3	
Solar Geoengineering			9.3		9.4		9.3 9.4			9.2
Tools for Individuals	10.1	10.1	10.2	10.3	10.2	10.1			10.2	10.2
Tools for Society		11.1	11.2 11.1	11.3	11.2 11.1	11.1 11.3	11.1	11	11.1	11.1 11.3
Education		12.2			12.1					
Finance		13.2	13				13.2			

Table 1: Climate change solution domains, along with areas of ML that are relevant to each. Rows of the table correspond to sections of this paper. This table should not be seen as comprehensive.

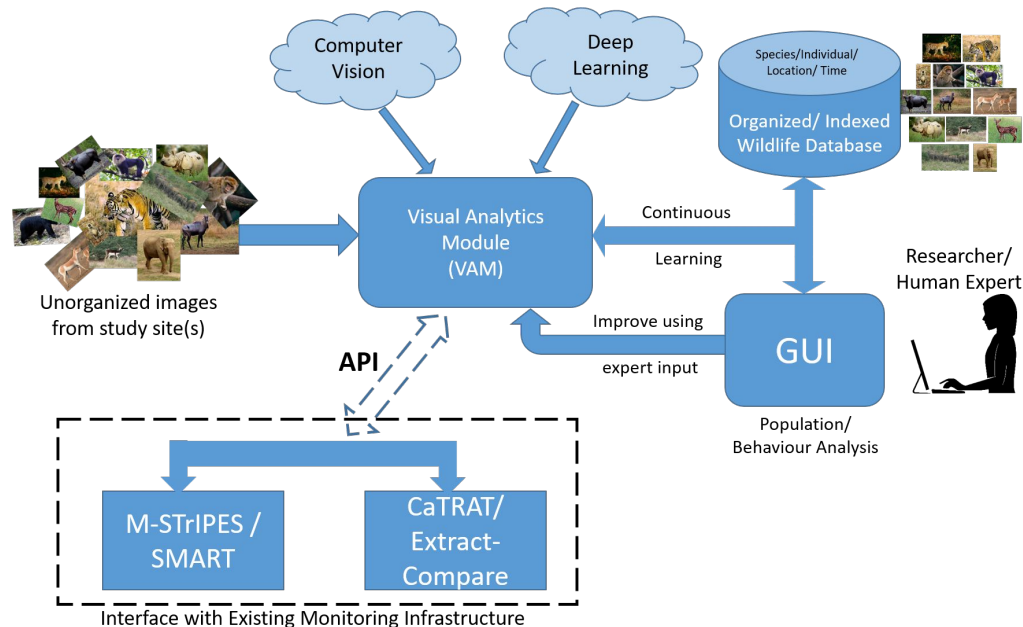
Source: Rolnick et al., “[Tackling Climate Change with Machine Learning](#)”, ArXiv, 10th Jun. 2019

- 22 authors from 16 organizations

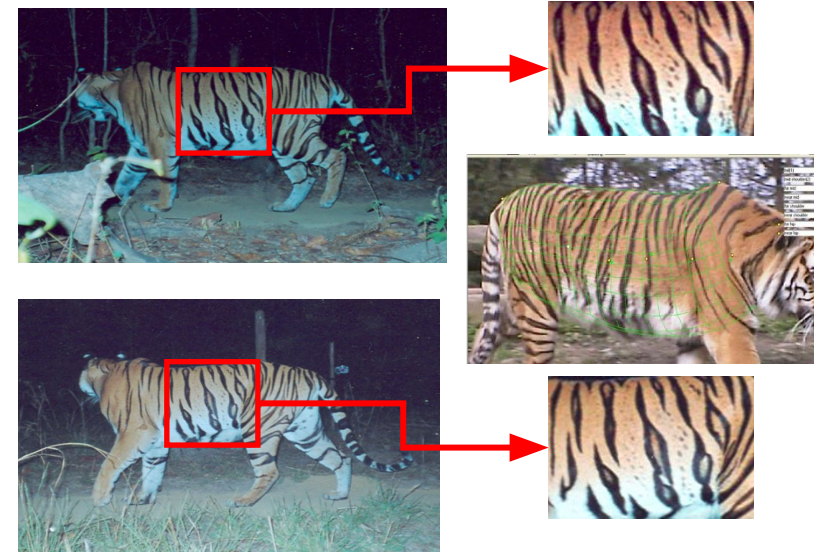
Vision for Wildlife

from conservation to conflict management

- **Intelligent Wildlife Monitoring**
 - **Goal:** automatic indexing of image datasets by species and individuals
 - **Applications:** camera-trap based population monitoring; crowd-sourced reporting of human- wildlife conflict



Individual Identification in Camera Trap Images



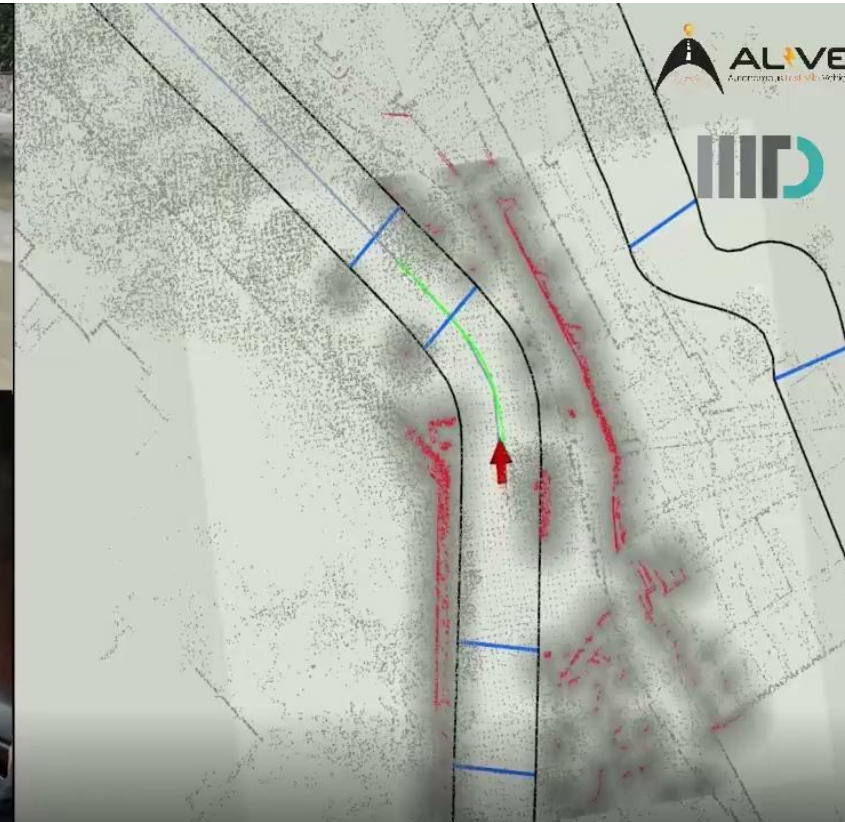
~31,000 images of ~1650 individual tigers ('13-'14): Over 70% tigers out of estimated population

ALIVE: Autonomous Driving @IIITD



- **Autonomous Last Mile Vehicle (ALIVE)**
 - Mahindra e2o & e-Rickshaw platforms
 - Campus testbed (outdoor)
 - Traffic Lights, Speed Bumps & Circles.
 - HD Maps, Real & Virtual Obstacles

Long-term Goal: Tackle the *urban Indian last mile*



विज्ञान एवं
प्रौद्योगिकी मंत्रालय
MINISTRY OF
SCIENCE AND
TECHNOLOGY

DST-PURSE

(Promotion of University
Research and Scientific
Excellence)



सत्यमेव जयते
Ministry of Electronics
& Information Technology
Government of India



New Vehicular Platform



- Migration to
 - Maini Golf Cart
 - ROS-2
- Still Work-in-Progress
 - Camera sensor mounting
 - Revamped multimodal localization



e-Cart with sensors



Rear panel for compute infra

Extended Reality (XR) Testbed



Spawning Virtual Obstacles

XR-Testbed: Generating Sensor Data



Photorealistic data stream generation using Deep Generative Models – GANs / Stable Diffusion Image from: “Learning to Segment via Cut-and-Paste”, ECCV 2018

What is Machine Learning?



- It is an important sub-area of AI that seeks to answer the following question:
How can we build computer systems that automatically improve with experience?
- More Formally:
“A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .” – Tom Mitchell

Machine Learning Tasks (T)



- Classification or Pattern Recognition
- Regression or Prediction
- Density Estimation
- Clustering
- Synthesis or Sampling
- Ranking
- Recommendation Systems
- Anomaly Detection/ Data Mining



Performance (P)



- A quantitative measure to evaluate performance
 - Usually Task specific
- Classification
 - Accuracy or Error Rate
 - Weighted versions of accuracy (e.g., in unbalanced data)
- Regression
 - Error measure such as ‘mean squared error’
 - Application dependent – many medium errors or few large errors?
- Density Estimation
 - log-probability assigned to examples

Almost always measured in terms of “generalization error”

Experience (E)



- Supervised Learning
 - Labelled data – (Data, target value)
 - Target value could be category/class labels, real value, real vector, etc.
 - Classification, Regression
- Unsupervised Learning
 - Only data, no labels
 - Density Estimation, Clustering
- Semi-supervised Learning
 - Some labelled data and lots of unlabelled data
 - Multiple-Instance Learning
- Reinforcement Learning
 - No examples, but a reward function
 - Payoff based on actions



Example Systems that use ML



- Web Search & Chatbots
 - ChatGPT, Gemini, Perplexity, Claude, DeepSeek, Sarvam...
- Amazon's recommendation system
- Adobe's Optical Character Recognition (OCR)
- Your phone's face tagging, photo organizer, low-light camera capabilities
- Instagram / Twitter feed
- Apple's Siri, Microsoft's Cortana, Amazon's Echo
- Youtube captions, your phone's automated captioning.
- Auto-parking and Advanced Driver Assistance Systems (ADAS)
- and many more ...

An Incomplete History of Learning

- Turing Test (1950)
 - Machines do very poorly
- Rosenblatt's Perceptron (1960's)
 - Kickstarted the mathematical analysis of the learning process
 - Key idea behind Support Vector Machines (SVMs) and Neural Networks
- Construction of Fundamentals of Learning Theory (1960-70's)
 - Focus on generalization capability of learning machines
 - Performance on unseen data
 - Regularization for ill-posed problems
 - e.g., linear equations for ill-conditioned matrices
- Neural Networks (1980's)
 - Connectionism
 - Back-propagation
 - CNNs, RNNs
- Learning Theory, SVMs (1990's)
 - Margin Maximization
 - Kernel Methods to handle non-linearity
- Deep Learning (>2006)
 - Hinton, Bengio, LeCun at forefront (Turing award '18)
 - Shift to DL-based solutions (>2012)
- Large Language / Multimodal / World Models (>2022)
 - Rise of OpenAI, Anthropic, DeepSeek, Sarvam, et al.

- Statistical Modelling and Inference
 - Regression, Hypothesis Testing, Significance Testing
- Signal Processing (Data = Signal)
 - Detection and Estimation
 - Hypothesis Testing (Classification/Detection)
 - Estimation (Regression – Error Minimization)
 - Representation of signals –
 - Fourier, Wavelets, etc.
 - Compressed sensing, Sparse representations, Dictionary learning.
- Optimization
 - Computational techniques for minimization/maximization of functions

What should you learn?



- Identifying and Modelling a Learning Problem
- Various algorithms (techniques) for solving ML problems
- Pitfalls while designing ML systems
 - Modelling, Generalization, Regularization & Model Selection, (hyper)-Parameter tuning, Overfitting, Underfitting
- Importance of Domain Knowledge
 - Not treating ML techniques as a black box
 - Simplify the learning problem by using domain knowledge
- ML Engineering Skills
 - Debugging ML models / systems to improve performance, reliability and trustworthiness.
- Tools
 - ML Libraries / packages – Scikit-learn, Pandas, PyTorch, TensorFlow, etc.
 - Visualization / Analytics – Seaborn, Plotly, Wandb, etc.

Course Outline

Topics:

- Empirical Risk Minimization Framework & Math Preliminaries
- Probabilistic Classifier (Bayes & Naïve Bayes)
- Linear Models for Regression and Classification
 - Linear & logistic regression
- ML in Practice
 - Training, Validation and Testing
 - Underfitting/Overfitting
 - Error Analysis
 - Hyperparameter search
 - Cross-validation
- Supervised Learning (beyond linear models)
 - Neural Networks
 - Loss and Optimization
 - Multilayer Perceptron
 - Support Vector Machines
 - Primal and Dual Formulations
 - Kernel Methods
 - Decision Trees and Random Decision Forests
- Unsupervised Learning
 - Clustering (K-means, DBSCAN, GMMs)
 - Dimensionality Reduction (PCA, LDA)

Prerequisites



● Required

- Linear Algebra – Vector spaces, linear independence, orthogonal bases, span, eigendecomposition, SVD.
- Probability and Statistics – Expectation, Variance, Covariance and their conditional variants, Bernoulli, Gaussians and their multivariate variants
- Vector Differentiation (Gradient, Jacobian)
- Introduction to Programming (Python)
 - In reality you would need much more than an introduction

● Desired

- Optimization
 - At least the notion of gradient descent used for function minimization

What's your advice for undergraduate student who aspires to be a research scientist in deep learning or related field one day?

3 Answers



Yann LeCun, Director of AI Research at Facebook and Professor at NYU

33.2k Views · Upvoted by Nikhil Dandekar, worked on machine learning at Microsoft, Foursquare and Quora and Tao Xu, Built ML systems at Airbnb, Quora, Facebook and Microsoft.

Most Viewed Writer in Machine Learning with 9 endorsements

- (o) take all the continuous math and physics class you can possibly take. If you have the choice between “iOS programming” and “quantum mechanics”, take “quantum mechanics”. In any case, take Calc I, Calc II, Calc III, Linear Algebra, Probability and Statistics, and as many physics courses as you can. But make sure you learn to program.

Learning Outcomes



Students are able to:

- Explain the different types of learning problems along with some techniques to solve them
- Model real-world problems, apply different learning techniques and quantitatively evaluate the performance
- Identify and use advanced techniques with the help of existing machine learning tools and libraries
- Analyze performance of ML techniques and comment on their limitations

- Course Webpage: <https://classroom.google.com/c/ODU1ODE0MTAwNTQ5?cjc=pnzakyya>
 - Access through my page as well
- Google Classroom Page (HW Submission/ Discussion): Machine Learning (ECE 363/563, CSE 343)
 - Class code: **pnzakyya** (also accessible through my page)
- Reading Material
 - Textbook & Lecture Notes posted on Google classroom
 - Machine Learning, Tom M. Mitchell, McGraw Hill, 1997
 - Understanding Machine Learning: From Theory to Algorithms, Shai Shalev-Shwartz and Shai Ben-David, Cambridge University Press, 2014
 - References* :
 - Machine Learning Yearning, Andrew Ng
 - Pattern Recognition and Machine Learning, Christopher M. Bishop, Springer, 2006
 - Pattern Classification, Richard Duda, Peter Hart and David Stork, 2nd ed., Wiley, 2006
 - Deep Learning, Ian Goodfellow, Aaron Courville, and Yoshua Bengio, 2016

Google Classroom



Google Classroom Code:

“pnzakyya”



<https://classroom.google.com/c/ODU1ODE0MTAwNTQ5?cjc=pnzakyya>

Tutorials & Resources



- Tutorials:

- Broader goal of developing the skills to debug, analyse, and improve ML model / system performance.
 - Python libraries – Numpy, Pandas, Sci-kit learn,
 - Visualization libraries for Exploratory Data Analytics – seaborn, matplotlib, plotly, etc.
 - Visualization of Math components – vectors, probability distributions, histograms, gradients for function optimization, etc.
 - Interactive plotting for data and error analysis

- Venue and Time:

- C101, Fridays 2:00PM – 4:00PM

- Resources:

- Will be posted regularly on Google Classroom

- Programming platform: Python
- Grading Scheme:
 - HW (4 Theory+Programming; individual) – 20%
 - Quiz (best 3/4) – 20% (No re-tests whatsoever)
 - Project & demo (in Groups) – 15%
 - Nature of projects (open-ended / Kaggle competitions / a combination?) will be determined based on final registration count.
 - Mid-sem – 20%
 - Final – 25%
- Relative Grading

Late Submission Policy



- All HW assignments are to be done and submitted individually
- Each student gets 3 “no-penalty extension” days over the semester.
 - They can be used all at once, or one at a time.
 - For any late submissions, extension days will automatically be applied.
 - The quantum of using extension days will be ‘one day’, even if the submission is late by the smallest unit of time.
 - Once the extension days are exhausted, any late submissions will not be evaluated and a zero will be awarded. Please use them wisely.
 - Extension days will not be applicable to group-work (project related submission deadlines).

Academic Integrity & Plagiarism policy



- Do NOT Plagiarize!
 - Do not take credit for work you did not do.
- Do NOT support Plagiarism!
 - Do not share your work with your peers until the evaluation is over.
- We will thoroughly check for plagiarism in your HW.
 - If you are caught plagiarizing, the institute's [academic dishonesty policy](#) will kick in.
 - No exceptions will be made regardless of delay in graduation or revocation of job or internship offers.
- In-class quiz / exams - **No smart devices - phones, watches, glasses, rings, etc.**
 - Dumb devices like good-old calculators are still permitted
 - Your responsibility to leave them away from your person
 - Failure to do so ***will*** result in a plag. charges, regardless of whether or not you were actively using it or if it was in your pocket.
- Following this rule is non-negotiable and you **MUST** submit your own work!

AI agent use in Assessment Components



- Using AI agents for Assignments is **Strongly Discouraged!**
 - HW Assignments are for learning - Using AI agents for coding & solving questions will adversely affect learning.
 - You may actively use AI agents for conceptual understanding.
 - If you still have to use AI agents for coding / solving questions in HW, please **do it responsibly** and justify and declare the usage.
 - Undeclared usage that matches with others will be considered as plagiarism
 - AI agents or not, you are expected to be able to explain your work in demos / viva / exams.
- Using AI agents for Projects is **Strongly Encouraged!**
 - Build the best things that you can using whatever resources you have access to.
 - Explain why what you built was the best that could be built, given the resources.

Next Class



- Review of Math Concepts –
 - Vector and Matrix Algebra
 - Multivariate Probability Distributions
 - Gradients and Jacobians
 - Optimization Problems
 - Learning Problems and the Empirical Risk Minimization Framework
-
- First Quiz (30-min.) on next Thursday, 20-Aug-2026
 - Linear Algebra, Probability & Statistics, Vector Differentiation

